

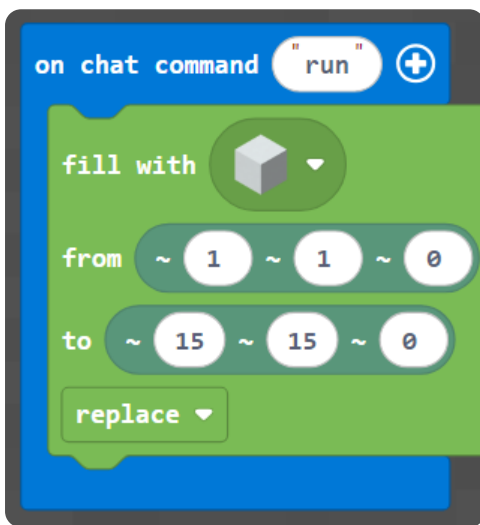
M003 Week 4 — English Worksheet (Beginner)

Topic: Pixel Art 2 — Cat & Tree (x, y) · **Course:** 3D Coordinates · **Level:** Beginner · **Time:** about 30 minutes

This week you copy a **cat** and a **tree**. Every block has a spot. The spot has **two numbers** — (x, y). x is how far **across**. y is how far **up**.

First, build your canvas. Run this to make a blank 15 × 15 wall, then put a **red block at your feet** as your **home spot**:

Blocks



Python

```
def on_run():
    blocks.fill(WHITE_CONCRETE,
                pos(1, 1, 0),
                pos(15, 15, 0),
                FillOperation.REPLACE)
    player.on_chat("run", on_run)
```

● **Stand on the red block every time you run.** If you move, the picture moves too. Same spot every time = a clean picture.

1 · Predict 🧠

Read the steps. Circle your answer. Do **not** run it yet.

```
place brown block at (8, 1)
place brown block at (8, 2)
place brown block at (8, 3)
```

Across or up? Circle one: across · up

Which number changes — x or y?

place orange block at (2, 15)

place orange block at (3, 15)

Across or up? Circle one: across · up

2 · Spot the Bug 🐛

This should make a **tree trunk** going **up** at $x = 8$: (8, 1), (8, 2), (8, 3). But all three blocks land in the **same spot**.

```
place brown block at (8, 1)
place brown block at (8, 1)
place brown block at (8, 1)
```

Hint: to go **up**, the **y** number must grow: 1, 2, 3.

Fill in the blanks to fix it:

```
place brown block at (8, 1)
place brown block at (8, __)
place brown block at (8, __)
```


Write one sentence: why was the first code wrong?

3 · Build It 📷

Stand on your **home spot** (the red block).

Warm up first. Copy this small tree. Read each square's (x, y) and place that color there.

🌳 Small tree

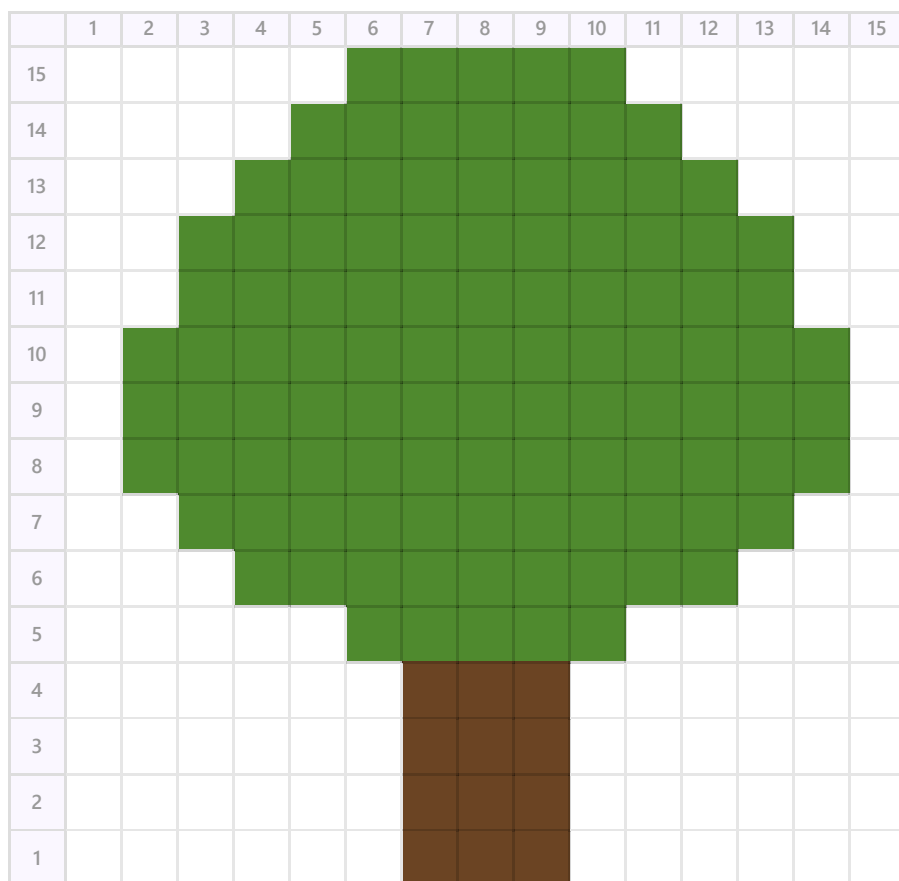
	1	2	3	4	5
5			🌿		
4		🌿	🌿	🌿	
3			🌳		
2			🌳		
1			🌳		

Now try a big one. Pick the 🐱 cat or the 🌳 tree below and copy it the same way. Read each colored square and place that color at its (x, y).

🐱 Cat face

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
15		🐱	🐱										🐱	🐱	
14		🐱	🐱	🐱								🐱	🐱	🐱	
13		🐱	🐱	🐱	🐱						🐱	🐱	🐱	🐱	
12		🐱	🐱	🐱	🐱	🐱	🐱	🐱	🐱	🐱	🐱	🐱	🐱	🐱	
11		🐱	🐱	🐱	🐱	🐱	🐱	🐱	🐱	🐱	🐱	🐱	🐱	🐱	
10		🐱	🐱	🐱	🐱	🐱	🐱	🐱	🐱	🐱	🐱	🐱	🐱	🐱	
9		🐱	🐱		🌿	🌿				🌿	🌿		🐱	🐱	
8		🐱	🐱	🐱	🐱	🐱	🐱	🐱	🐱	🐱	🐱	🐱	🐱	🐱	
7		🐱	🐱	🐱	🐱	🐱		👁		🐱	🐱	🐱	🐱	🐱	
6		🐱	🐱	🐱	🐱	🐱		👃		🐱	🐱	🐱	🐱	🐱	
5		🐱	🐱	🐱	🐱	🐱	🐱	🐱	🐱	🐱	🐱	🐱	🐱	🐱	
4			🐱	🐱	🐱	🐱	🐱	🐱	🐱	🐱	🐱	🐱	🐱	🐱	
3			🐱	🐱	🐱	🐱	🐱	🐱	🐱	🐱	🐱	🐱	🐱	🐱	
2				🐱	🐱	🐱	🐱	🐱	🐱	🐱	🐱	🐱	🐱		
1						🐱	🐱	🐱	🐱	🐱					

 Tree



When you finish, send a photo. Then write 2 or 3 sentences.

I built the ...

The lowest block is at $(x = \dots, y = \dots)$.

4 · Explain It 🎥

Take a video on a phone. Show your picture and talk like you are teaching a friend. Try to use these words: **x, y, across, up, home spot**.

1. Show your home spot (the red block).
2. Show your finished picture.
3. Point at one block and say its spot: "this one is at x ..., y ...".

Now write 2 short sentences about your picture. You can read them in your video.

Submit ✓

Send this worksheet + your walkthrough video to teacher on KakaoTalk.